Mathletes Formulas

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1. Polynomials

1.1. Linear Equations

Slope-Intercept $y = mx + b$ **Slope** $m = \dfrac{\Delta y}{\Delta x} = \dfrac{y_2 - y_1}{x_2 - x_1}$

Point-Slope $y - y_1 = m(x - x_1)$ **Point-Point** $y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Parallel lines have equivalent slopes, while perpendicular lines have slopes that are negative reciprocals of one another ($m \perp -\frac{1}{m}$)

1.2. Quadratics

Quadratic Equations

Standard Form $ax^2 + bx + c$

Vertex Form $a(x - h)^2 + k$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Let $D = b^2 - 4ac$,

1. $D > 0 \implies$ 2 real solutions
2. $D = 0 \implies$ 1 real solution
3. $D < 0 \implies$ no real solutions

Minimum and Maximum Values

Consider the function $f(x) = ax^2 + bx + c$ with vertex $\left(-\frac{b}{2}a, f\left(-\frac{b}{2}a\right)\right)$

1. $a > 0 \implies f$ has a *minimum* at $x = -\frac{b}{2}a$. The minimum value is $f\left(-\frac{b}{2}a\right)$.
2. $a < 0 \implies f$ has a *maximum* at $x = -\frac{b}{2}a$. The maximum value is $f\left(-\frac{b}{2}a\right)$.

1.3. General Polynomials

Let n be a non-negative integer and let $a_n, a_{n-1}, ..., a_2, a_1, a_0$ be real numbers such that $a_n \neq 0$. The function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_2 x^2 + a_1 x + a_0$$

is a polynomial function of x with degree n.

Linear Factorization Theorem

If $f(x)$ is a polynomial of degree n, where $n > 0$, then $f(x)$ has precisely n linear factors

$$f(x) = a_n(x - c_1)(x - c_2)...(x - c_n)$$

where $c_1, c_2, ..., c_n$ are complex numbers.

Rational Zero Test

If the polynomial has *integer* coefficients, then every rational zero of f has the form $\frac{p}{q}$, where p and q have no commons factors other than 1, and

$p =$ a factor of the constant term a_0 $q =$ a factor of the leading coefficient a_n

Conjugate Pairs

Let f be a polynomial function with *real coefficients*. If $a + bi$, where $b \neq 0$, is zero of f, the complex conjugate $a - bi$ is also a zero of f.

For polynomials with *integer coefficients*, $a + \sqrt{b}$ being a zero implies $a - \sqrt{b}$ is also a zero.

Roots

Let f be the polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_2 x^2 + a_1 x + a_0$$

The sum of all roots $= -\dfrac{a_n}{a_{n-1}}$ The product of all roots $= (-1)^n \dfrac{a_0}{a_n}$

Binomial Theorem

Let n be a non-negative integer

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k, \text{ where } \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

1.4. Rational Functions

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + ... + b_1 x + b_0}$$

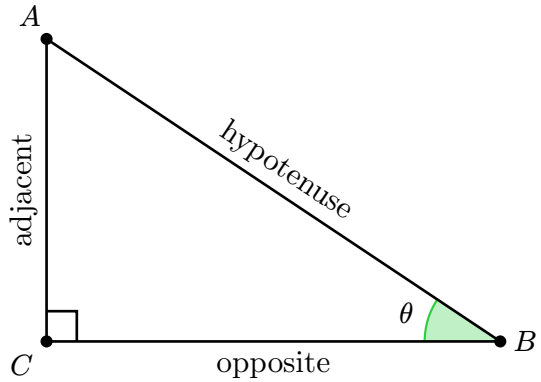
where $N(x)$ and $D(x)$ have no common factors.

1. The graph of f has *vertical* asymptotes at the zeros of $D(x)$.
2. The graph of f has at most one *horizontal asymptote* determined by comparing the degrees of $N(x)$ and $D(x)$.
    a. $n < m \implies$ the line $y = 0$ is the horizontal asymptote.
    b. $n = m \implies$ the line $y = \frac{a_n}{b_m}$ is the horizontal asymptote.
    c. $n > m \implies$ there is no horizontal asymptote.

Suppose $N(x)$ and $D(x)$ had some common factor $(x - c)$. The graph of f would then have a hole at $(c, f(c))$

2. Trigonometry

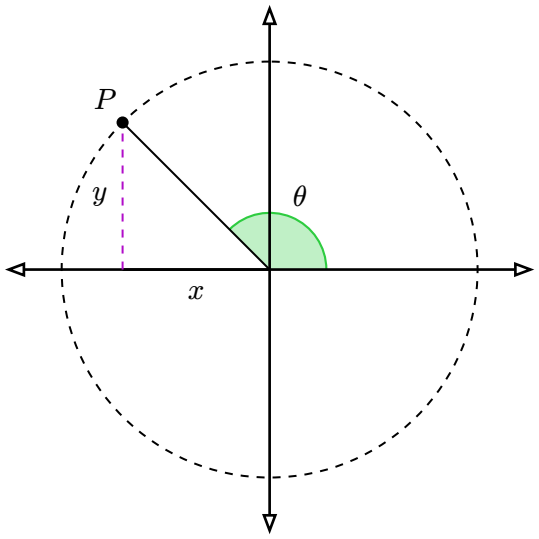
2.1. Functions



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} \qquad \cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$



$$\sin\theta = \frac{y}{1} = y \qquad \csc\theta = \frac{1}{y}$$

$$\cos\theta = \frac{x}{1} = x \qquad \sec\theta = \frac{1}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{x}{y}$$

Function	Domain	Range	Period
sin	$\mathbb{R}$	$[-1, 1]$	2π
cos	$\mathbb{R}$	$[-1, 1]$	2π
tan	$\left\{x \mid x \in \mathbb{R}, x \neq \frac{\pi}{2} + n\pi\right\}$ *	$\mathbb{R}$	π
csc	$\{x \mid x \in \mathbb{R}, x \neq n\pi\}$ *	$(-\infty, -1] \cup [1, \infty)$	2π
sec	$\left\{x \mid x \in \mathbb{R}, \pi \neq \frac{\pi}{2} + n\pi\right\}$ *	$(-\infty, -1] \cup [1, \infty)$	2π
cot	$\{x \mid x \in \mathbb{R}, x \neq n\pi\}$ *	$\mathbb{R}$	π

Table 1: Functions (* $\forall n \in \mathbb{Z}$)

Function	Domain	Range
$\sin^{-1}$ / arcsin	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$ / arccos	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$ / arctan	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\csc^{-1}$ / arccsc	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
$\sec^{-1}$ / arcsec	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
$\cot^{-1}$ / arccot	$\mathbb{R}$	$(0, \pi)$

Table 2: Inverse Functions

	0°	**30°**	**45°**	**60°**	**90°**	**180°**	**270°**	**360°**
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1
cos	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	**undef**	0	**undef**	0

Table 3: Common Values

	QI	**QII**	**QIII**	**QIV**
sin / csc	(+)	(+)	(−)	(−)
cos / sec	(+)	(−)	(−)	(+)
tan / cot	(+)	(−)	(+)	(−)

Table 4: Function sign values in different quadrants

You can use the mnemonic “All Students Take Calculus” (ASTC) to remember that in

QI. **A**ll functions are positive QII. Only **S**ine is positive

QIII. Only **T**angent is positive QIV. Only **C**osine is positive

2.2. Identities

Radians and Degrees

$$n^\circ = \frac{\pi}{180} n \text{ rad}$$
$$m \text{ rad} = \left(\frac{180}{\pi} m\right)^\circ$$

Reciprocals

$$\csc\theta = (\sin\theta)^{-1} \quad \sin\theta = (\csc\theta)^{-1}$$
$$\sec\theta = (\cos\theta)^{-1} \quad \cos\theta = (\sec\theta)^{-1}$$
$$\cot\theta = (\tan\theta)^{-1} \quad \tan\theta = (\cot\theta)^{-1}$$

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$
$$\tan^2\theta + 1 = \sec^2\theta$$
$$1 + \cot^2\theta = \csc^2\theta$$

Even/Odd

$$\sin(-\theta) = -\sin\theta \quad \csc(-\theta) = -\csc\theta$$
$$\cos(-\theta) = \cos\theta \quad \sec(-\theta) = \sec\theta$$
$$\tan(-\theta) = -\tan\theta \quad \cot(-\theta) = -\cot\theta$$

Cofunction

$$\sin(90^\circ - \theta) = \cos\theta \quad \cos(90^\circ - \theta) = \sin\theta$$
$$\sec(90^\circ - \theta) = \csc\theta \quad \csc(90^\circ - \theta) = \sec\theta$$
$$\tan(90^\circ - \theta) = \cot\theta \quad \cot(90^\circ - \theta) = \tan\theta$$

Sum and Difference

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

Double Angle

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2 x$$
$$= 2\cos^2\theta - 1$$
$$= 1 - 2\sin^2\theta$$
$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

Sum to Product

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\sin\alpha - \sin\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$
$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

Half Angle

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$
$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$
$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta}$$

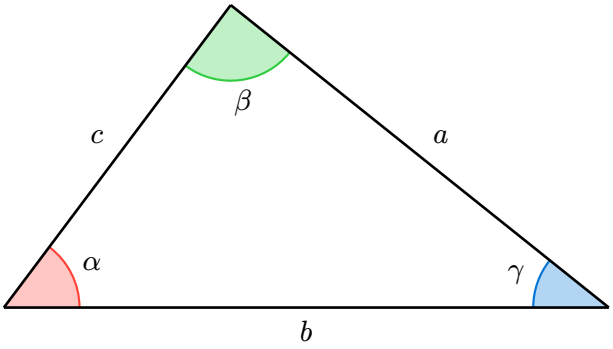
Product to Sum

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta) - \cos(\alpha+\beta)}{2}$$
$$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta) + \cos(\alpha-\beta)}{2}$$
$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta) + \sin(\alpha-\beta)}{2}$$
$$\cos\alpha\sin\beta = \frac{\sin(\alpha+\beta) - \sin(\alpha-\beta)}{2}$$

Power Reduction

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$
$$\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$$

2.3. Laws



$$\text{Area} = \frac{1}{2}bc\sin\alpha = \frac{1}{2}ac\sin\beta = \frac{1}{2}ab\sin\gamma$$

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc\cos\alpha$$
$$b^2 = a^2 + c^2 - 2ac\cos\beta$$
$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$
$$\cos\alpha = \frac{b^2 + c^2 - a^2}{2bc}$$
$$\cos\beta = \frac{a^2 + c^2 - b^2}{2ac}$$
$$\cos\gamma = \frac{a^2 + b^2 - c^2}{2ab}$$

Law of Sines

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$$
$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

Mollweide's Formula

$$\frac{a+b}{c} = \frac{\cos\left(\frac{1}{2}(\alpha-\beta)\right)}{\sin\left(\frac{1}{2}\gamma\right)}$$

Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{1}{2}(\alpha-\beta)\right)}{\tan\left(\frac{1}{2}(\alpha+\beta)\right)}$$
$$\frac{b-c}{b+c} = \frac{\tan\left(\frac{1}{2}(\beta-\gamma)\right)}{\tan\left(\frac{1}{2}(\beta+\gamma)\right)}$$
$$\frac{a-c}{a+c} = \frac{\tan\left(\frac{1}{2}(\alpha-\gamma)\right)}{\tan\left(\frac{1}{2}(\alpha+\gamma)\right)}$$

3. Graphs and Geometry

3.1. Lines and Points

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Pythagorean Theorem

$$a^2 + b^2 = c^2 \implies c = \sqrt{a^2 + b^2}$$

Perpendicular Bisector

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = x_2^2 + y_2^2 - x_1^2 - y_1^2$$

3.2. Circles

$$(x-m)^2 + (y-n)^2 = r^2$$

$$\text{Area} = \pi r^2$$

$$\text{Circumference} = 2\pi r$$

3.3. Triangles

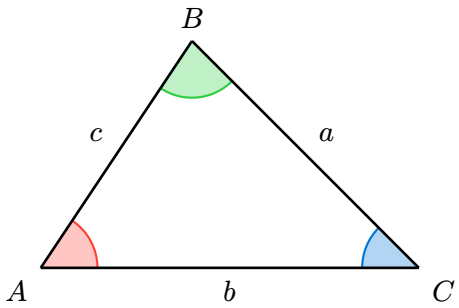
Similarity

Two triangles $\triangle ABC$ and $\triangle DEF$ are similar if

SSS	SAS	AA
$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$	$\frac{AB}{DE} = \frac{AC}{DF}$ and $m\angle A = m\angle D$	$m\angle A = m\angle D$ $m\angle B = m\angle E$

Oblique Trangles

Oblique triangles have no right angles. To solve an oblique triangle, you need to know the measure of at least one side along with any two other measures.



1. Two angles and a side (AAS or ASA)
2. Two sides and an opposites angle (SSA)
3. Three sides (SSS)
4. Two sides and their included angle (SAS)

The first two cases can be solved using the **Law of Sines**, whereas the last two require the **Law of Cosines**

Ambiguous Case (SSA)

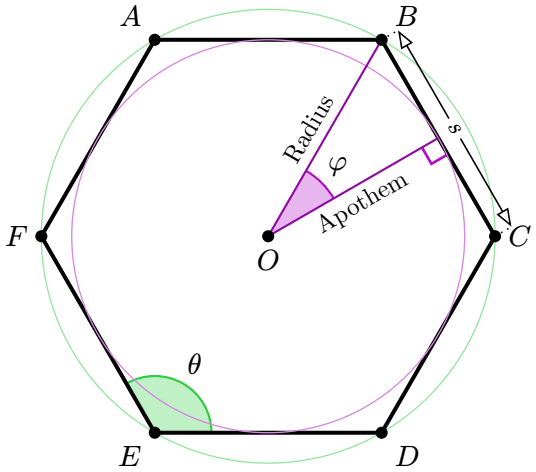
Consider a triangle in which $a, b,$ and A are given. $(h = b \sin A)$

A	**acute**	**acute**	**acute**	**acute**	**obtuse**	**obtuse**
Condition	$a < b$	$a = h$	$a \geq b$	$h < a < b$	$a \leq b$	$a > b$
Trangles Possible	**0**	**1**	**1**	**2**	**0**	**1**

Heron's Area Formula

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}, \text{where } s = \frac{a+b+c}{2}$$

3.4. Regular Polygons



$$\theta = \frac{180(n-2)^\circ}{n}$$

$$\varphi = \left(\frac{180^\circ}{2n}\right)$$

$$\text{Apothem} = r\cos\varphi$$

$$\text{Side Length} = 2r\sin\varphi$$

$$\text{Area} = \frac{1}{2}\cdot \text{Apothem} \cdot \text{Perimeter}$$

$$\sum \text{Interior Angles} = 180(n-2)^\circ$$

3.5. Volumes

$$V_{\text{Cylinder}} = \pi r^2 h \qquad V_{\text{Cone}} = \frac{1}{3}\pi r^2 h \qquad V_{\text{Sphere}} = \frac{4}{3}\pi r^3 \qquad V_{\text{Pyramid}} = \frac{1}{3}Bh$$

3.6. Graph Manipulation

Transformation	Conditions	Description
$g(x) = f(x \pm c)$	$(+)$ $(-)$	shift left c units shift right c units
$g(x) = f(x) \pm c$	$(+)$ $(-)$	shift left c units shift right c units
$g(x) = cf(x)$	$c < 1$ $c > 1$	vertical compress by a factor of c vertical stretch by a factor of c
$g(x) = f(cx)$	$c < 1$ $c > 1$	horizontal stretch by a factor of c horizontal compress by a factor of c
$g(x) = -f(x)$	**N/A**	reflect about the x-axis
$g(x) = f(-x)$	**N/A**	reflect about the y-axis

4. Miscellaneous

4.1. Inverse Functions

Let f and g be two function such that

$f(g(x)) = x$ for all x in the domain of g $\qquad$ $g(f(x)) = x$ for all x in the domain of f

Under these conditions, the function g is the *inverse function* of the function f (and vice-versa). The function g can then be denoted as f^{-1}. So,

$$f(f^{-1}(x)) = x \qquad f^{-1}(f(x)) = x$$

The domain of f must be equal to the range of f^{-1}, and the range of f must be equal to the domain of f^{-1}.

4.2. Series

Arithmetic

$$S_n = \frac{n}{2}(a_1 + a_n)$$

$$a_n = a_1 + (n-1)d$$

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Geometric

$$S_n = \frac{a_1(1-r^n)}{1-r}, r \neq 0$$

$$\sum_{i=1}^{\infty} a_i = \frac{a_1}{1-r}, |r| < 1 *$$

$$a_n = a_1 r^{n-1}$$

* where there are infinitely many a terms

4.3. Complex Numbers

Let a and b be real numbers. The number $a + bi$ is a *complex number* written in *standard form*. The number a is the *real part* and the number bi is the *imaginary part* of the complex number. The absolute value of an imaginary number is defined as its distance from the origin, $\sqrt{a^2 + b^2}$

When $b = 0$, the number $a + bi$ is a real number. When $b \neq 0$, the number $a + bi$ is an *imaginary number*. A number of the form bi, where $b \neq 0$, is a *pure imaginary number*.

$$i^0 = 1, i^1 = i, i^2 = -1, i^3 = -i, i^n = i^{n \bmod 4}$$

Trigonometric Form

$$z = a + bi = r(\cos\theta + i\sin\theta)$$

such that $r = |z| = \sqrt{a^2 + b^2}$, $\tan\theta = \frac{b}{a}$, $a = r\cos\theta$ and $b = r\sin\theta$, Here r is the *modulus* of z, and θ is an *argument* of z.

Product and Quotient

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ be complex numbers

$$z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)) \qquad \frac{z_1}{z_2} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)), z_2 \neq 0$$

DeMoivre's Theorem

If $z = r(\cos\theta + i\sin\theta)$ is a complex number and n is positive integer, then

$$z^n = (r(\cos\theta + i\sin\theta))^2 = r^n(\cos(n\theta) + i\sin(n\theta))$$

Roots of Unity

For a positive integer n, the complex number $z = r(\cos\theta + i\sin\theta)$ has exactly n distinct nth roots given by

$$z_k = \sqrt[n]{r}\left(\cos\frac{\theta+2\pi k}{n} + i\sin\frac{\theta+2\pi k}{n}\right),\ k = 0, 1, 2, ..., n-1$$

In other words, $x^n = a + bi \Rightarrow x = z_0, z_1, z_2, ...z_{n-1}$

4.4. Systems of Equations

Method of Substitution

1. *Solve* one of the equations for one variable in terms of the other.
2. *Substitute* the previous expression into the other equation.
3. *Solve* the equation obtained in Step 2
4. *Back-substitute* the value obtained in Step 3 into the expression obtained in Step 1 to find the value of the other variable
5. *Check* that the solution satisfies each of the original equations

Method of Elimination

1. *Obtain coefficients* for x (or y) that differ only in sign.
2. *Add* the equations to eliminate one variable

Guassian Elimination

Each of the following *row operations* on system of linear equations produces an *equivalent* system of equations. The goal is to reache *Row-Echelon Form.*

1. Interchange two equations
2. Multiply one of the equations by a nonzero constant
3. Add a multiple of one equation to another equation to replace the latter equation

Row-Echelon Form

$$\begin{cases} x + a_1y + a_2z = b_1 \\ y + a_3z = b_2 \\ z = b_3 \end{cases}$$

4.5. Exponential and Logarithmic Properties

Exponential

$$a^0 = 1 \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$
$$a^m \cdot a^n = a^{m+n} \qquad (a^m)^n = a^{m\cdot n}$$
$$\frac{a^m}{a^n} = a^{m-n} \qquad a^{\frac{1}{n}} = \sqrt[n]{a}$$
$$(ab)^n = a^n \cdot b^n \qquad a^m = a^n \Longrightarrow m = n$$

Logarithmic

$$\log_n 1 = 0$$
$$\log_n n = 1$$
$$\log x = \log_{10} x$$
$$\ln x = \log_e x$$
$$n^{\log_n x} = x$$
$$\log_n n^x = x$$
$$\log_n a = b \implies a = n^b$$

$$\log_n a^b = b \log_n a$$
$$\log_{n^b} a = \frac{1}{b} \log_n a$$
$$\log_n ab = \log_n a + \log_n b$$
$$\log_n \frac{a}{b} = \log_n a - \log_n b$$
$$\log_n a = \log_n b \implies a = b$$